

The SDPNRI Compartmental Model of Rumour Spread

The influence of emotion contagion and political bias on debunkers threshold

1 Introduction

Epidemiological modeling frameworks, such as the classical Susceptible-Infectious-Recovered (SIR) model, have long provided foundational mathematical tools for understanding how information spreads through populations. While these methods originated in public health, researchers have increasingly adapted compartmental dynamics to study the diffusion of digital information and misinformation across online networks. My motivation to build this project began when I critically questioned the extent to which existing frameworks, such as the recent SEDPNR model <https://www.nature.com/articles/s41598-024-69657-0> (Susceptible, Exposed, Doubtful, Positively-Infected, Negatively-Infected, Restrained), truly represent real-world sociological dynamics. Specifically, traditional models often assume that agents cannot gain complete immunity and will inevitably fall back into a susceptible state. Furthermore, models incorporating an "exposed" (E) compartment treat exposure as a latent phase akin to physical disease incubation. However, in digital environments, exposure to misinformation is instantaneous, and standard frameworks lack compartments that capture the active sociological impact of debunkers and experts. To address these limitations, this project introduces a custom agent-based SDPNRI model. By removing the exposed compartment and introducing an Immune (I) compartment representing active debunkers, this framework investigates a core research question: How does the introduction of political confirmation bias change the macroscopic density of active debunkers required to contain an online rumor compared to a purely emotional baseline? Using dynamic retweet probabilities where political bias acts as a multiplier against emotional intensity, the study evaluates how automated institutional intervention strategies can be scaled to maintain containment across polarized social networks.

2 Methodology

2.1 The SDPNRI Compartmental Model

For this research paper, an SDPNRI compartmental model was adopted to represent different states for social media users (Susceptible, Doubtful, Positively-infected, Negatively-infected, Restrained, Immune). This framework extends classical epidemic modeling to capture the dynamics of rumour spread.

In social networks, the force of infection represents the per-capita rate at which an individual encounters rumour-spreading content. Unlike real pathogens, which could spread faster with a scaling susceptible density in a physical environment, misinformation spread doesn't depend on the number of users but rather on network topology, such as the number of followers and feed algorithms; intuitively, having a bigger population doesn't imply having more social connections than in a smaller one. Therefore, my model is frequency dependent. (book)

In classical epidemiology frameworks, we assume homogeneous mixing and passive hosts, where all individuals share identical properties and equal probabilities of transitioning from one state to another. However, rumor dynamics are strongly influenced by cognitive biases, polarities, and emotions. SDPNRI extends

classic SIR models by adopting behavioural and psychological phenomena that drive real-world social networks dynamics.

2.2 Model Compartments

Doubtful Represents individuals who, upon exposure, question the reliability of the information and express uncertainty until they either lose interest, commit to a polarised stance, or gain access to fact-checking sources.

Infection (P/N)

Physical pathogens typically infect hosts uniformly, while infection in social networks triggers different ideologies and affective responses. Positively-Infected users propagate the rumours with positive sentiments and endorsement, while Negatively-Infected users circulate the same rumour with negative emotions, outrage, and criticism. Recent studies show that negative news tends to spread faster than positive ones, which justifies this modeling decision.

Restrained Represents people who, over time, lose interest in the rumour and stop spreading it or questioning it because of social-media fatigue.

Immune Represents people with enough expertise to know that a rumour is false, or fact-checkers who consult reliable sources. The Immune state is absorbing, rendering all subsequent state transitions impossible.

2.3 State Transition Diagram

Figure 1 shows the directed state-transition graph for the SDPNRI model.

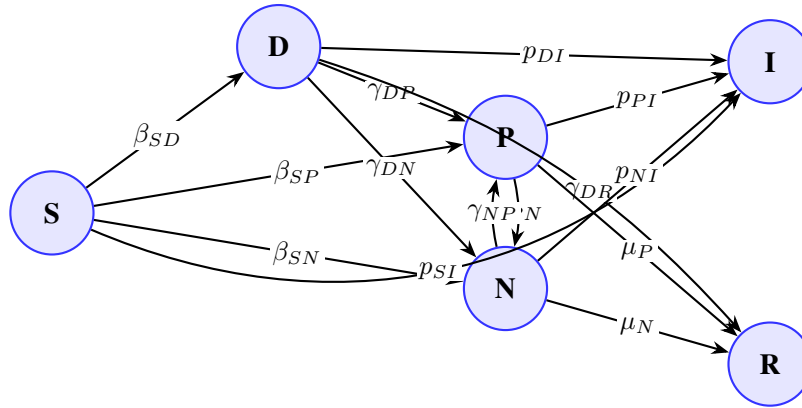


Figure 1: Directed state transition graph for the SDPNRI model.

2.4 Governing Equations

The system of ordinary differential equations describing the macro-level SDPNRI dynamics is given by:

$$\begin{aligned}
\frac{dS}{dt} &= -p_{SI}S - (\beta_{SD} + \beta_{SP} + \beta_{SN})S(P + N) \\
\frac{dD}{dt} &= \beta_{SD}S(P + N) - (\gamma_{DP} + \gamma_{DN} + \gamma_{DR} + p_{DI})D \\
\frac{dP}{dt} &= \beta_{SP}S(P + N) + \gamma_{DP}D - \mu_P P - p_{PI}P - \gamma_{PN}P + \gamma_{NP}N \\
\frac{dN}{dt} &= \beta_{SN}S(P + N) + \gamma_{DN}D - \mu_N N - p_{NI}N - \gamma_{NP}N + \gamma_{PN}P \\
\frac{dI}{dt} &= p_{SI}S + p_{DI}D + p_{PI}P + p_{NI}N \\
\frac{dR}{dt} &= \gamma_{DR}D + \mu_P P + \mu_N N
\end{aligned}$$

Table 1: SDPNRI Model Transition Rates

Rate	Transition
p_{SI}	$S \rightarrow I$
β_{SD}	$S \rightarrow D$
β_{SP}	$S \rightarrow P$
β_{SN}	$S \rightarrow N$
γ_{DP}	$D \rightarrow P$
γ_{DN}	$D \rightarrow N$
γ_{DR}	$D \rightarrow R$
p_{DI}	$D \rightarrow I$
p_{PI}	$P \rightarrow I$
p_{NI}	$N \rightarrow I$
μ_P	$P \rightarrow R$
μ_N	$N \rightarrow R$

The system of ordinary differential equations presented above describes the macro-level theoretical framework of the SDPNRI model in a homogenous approach, assuming continuous population-level transmission rates. Therefore, to test and validate this theoretical model against real-world social network dynamics – including asynchronous user activities, political biases, and emotional contagion – an Agent-Based Model (ABM) was created, calibrated using a real dataset.

2.5 Dataset

For this research paper, a subset of the PHEME-Aug (v1.0) augmented rumor dataset published by Han, Gao, and Ciravegna (2019) was used. This corpus enhances the original PHEME dataset for rumor detection (Kochkina et al., 2018) by adding supplementary contextual threads from the CrisisLexT26 crisis repository (Olteanu et al., 2015).

While PHEME-Aug covers multiple historical events, this study explicitly isolates the sub-corpus tracking the 2013 Boston Marathon bombings only, specifically the rumour that claimed the death of an 8 year-old girl at the Boston Marathon finish line while running to raise money for the victims of the Sandy Hook Elementary School shooting (which had occurred four months prior, in December 2012). The original story

involved an 8 year old boy named Martin Richard who was killed while cheering for his family and friends near the finish line.

This data is extracted from Twitter, which was renamed “X” in 2023. It is a microblogging platform that allows users to post personal content such as tweets, and navigate a feed page driven by algorithms that prioritize recent content from people they follow. It offers options to ‘retweet’ posts, which would increase its reach and spread it to the retweeter’s network, or react to it through comments and likes. My chosen dataset tracks twitter activities at the 2013 Boston bombings event, and it includes text content, posting time, user_id... My ABM will simulate a closed Twitter population ruled by transmission rates concluded from the calibration of my dataset.

2.6 Calibration of Transmission Rates

To calibrate my transmission rates, I developed a computer program that extracts the text across all activities in my dataset and feeds it to Claude AI to classify into Infectious, Immune, Doubtful, and Irrelevant, using a well structured prompt where Irrelevant represents activities that do not relate to the rumour topic. For Infectious categories, VADER sentiment-analysis was used to classify the infectious activities into Positively-Infected or Negatively-Infected based on the tonality of the text. VADER is a lexicon-based tool designed to assign a compound score between -1 (most extreme negative) and $+1$ (most extreme positive). Scores above 0.05 mean positive, below -0.05 mean negative, and values in between mean neutral.

For the purpose of this study, neutral sentiment classification was omitted, and the new threshold for negative classification was adjusted to below -0.3 , which was tested empirically, due to the high negative emotions that revolve around a child’s death rumour.

Because my data can not specify the exact moment a user becomes inactive, a 12-hour duration of continuous inactivity following an agent’s last recorded post was operationalized as the transition threshold into the Restrained (R) compartment. Recent studies demonstrate that over 60% of crisis-related retweeting cascades naturally cease diffusion within a 12-hour window as social media attention shifts and information fatigue sets in (Li et al., 2021; Pfeffer et al., 2023).

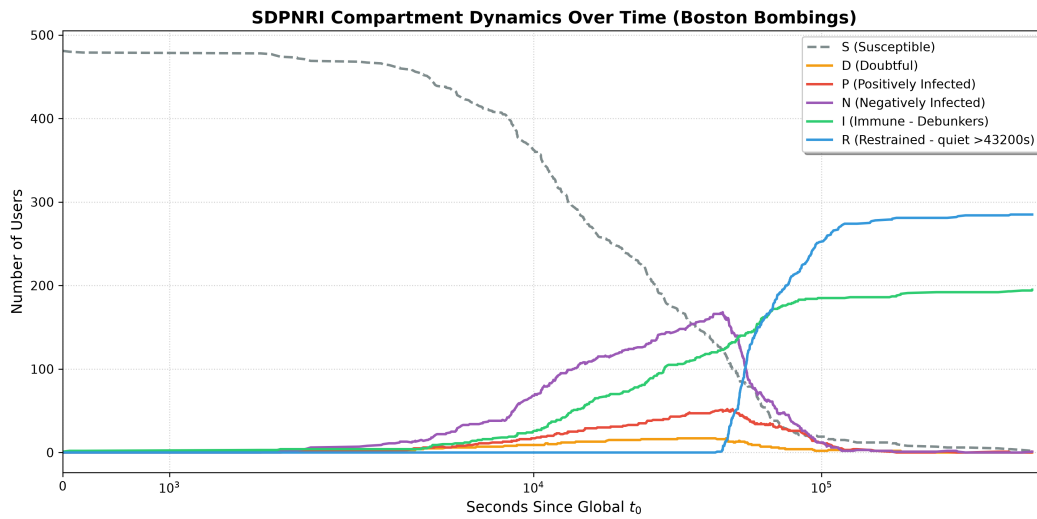


Figure 2: SDPNRI compartment dynamics over time, calibrated on the Boston Marathon bombings dataset. $N = 482$.

The model updates compartment populations dynamically as a function of elapsed time t (measured in seconds since global t_0), as shown in Figure 2. Because individual agent activities are asynchronous, state

transitions occur at discrete empirical timestamps across the total population $N = 482$.

To calibrate the model transition parameters, empirical hazard rates were estimated using the person-time incidence rate method. For each permitted transition from A to B , the rate per second was derived by dividing the total count of observed transitions by the cumulative person-time exposure spent by agents in the source state A before transitioning to another state.

$$\lambda_{A \rightarrow B} = \frac{N_{A \rightarrow B}}{\text{Exposure}_A} = \frac{N_{A \rightarrow B}}{\sum_{i=1}^M A(t_i) \cdot \Delta t_i}$$

Table 2: Empirically Calibrated Transition Rates

Transition	Events	Person-seconds	Rate per second
$D \rightarrow I$	2	1,229,508	1.6267×10^{-6}
$D \rightarrow N$	0	1,229,508	0
$D \rightarrow P$	2	1,229,508	1.6267×10^{-6}
$N \rightarrow D$	5	8,697,323	5.7489×10^{-7}
$N \rightarrow I$	3	8,697,323	3.4493×10^{-7}
$N \rightarrow P$	16	8,697,323	1.8396×10^{-6}
$P \rightarrow D$	1	3,508,268	2.8504×10^{-7}
$P \rightarrow I$	2	3,508,268	5.7008×10^{-7}
$P \rightarrow N$	2	3,508,268	5.7008×10^{-7}
$S \rightarrow D$	16	17,306,130	9.2453×10^{-7}
$S \rightarrow I$	188	17,306,130	1.0863×10^{-5}
$S \rightarrow N$	214	17,306,130	1.2366×10^{-5}
$S \rightarrow P$	64	17,306,130	3.6981×10^{-6}

2.7 Agent-Based Model

Using the calibrated transmission rates above, I developed an agent based model with the same exact properties of my previous population, and I received the results shown in Figure 3. The close alignment between both diagrams reflects the reliability of person-time incidence rates in generating transmission rates that mimic the real-world social networks dynamics.

Rumour spread on social media depends on many factors, such as follower count, activity levels, the age of the post, and the attractiveness of the post, which we assume are all embedded within our calibrated transmission rates. However, to address my research question regarding the effect of emotional contagion and political bias in the immune compartment thresholds, adding a political bias is crucial to my ABM, as the child’s death rumour barely reflects any political bias but is rather heavily driven by emotions. Political bias, in return, amplifies the emotion, whether through outrage and criticism or endorsement and support. Therefore, I developed a theoretical multiplier for the $S \rightarrow P$ and $S \rightarrow N$ transitions that shows how the transmission rates will evolve and consequently affect other compartments as well.

2.8 Political Bias and Emotional Contagion Extension

The baseline emotional score of the post (E_{base} , derived from the magnitude of the compound VADER score $|S_{\text{vader}}|$) is amplified by the agent’s political intensity, calculated as its distance from political neutrality (Bias = 1.0). To account for the well-established bimodal nature of affective polarization, where emotional arousal and transmission susceptibility remain moderate near neutral stances but intensify toward ideological

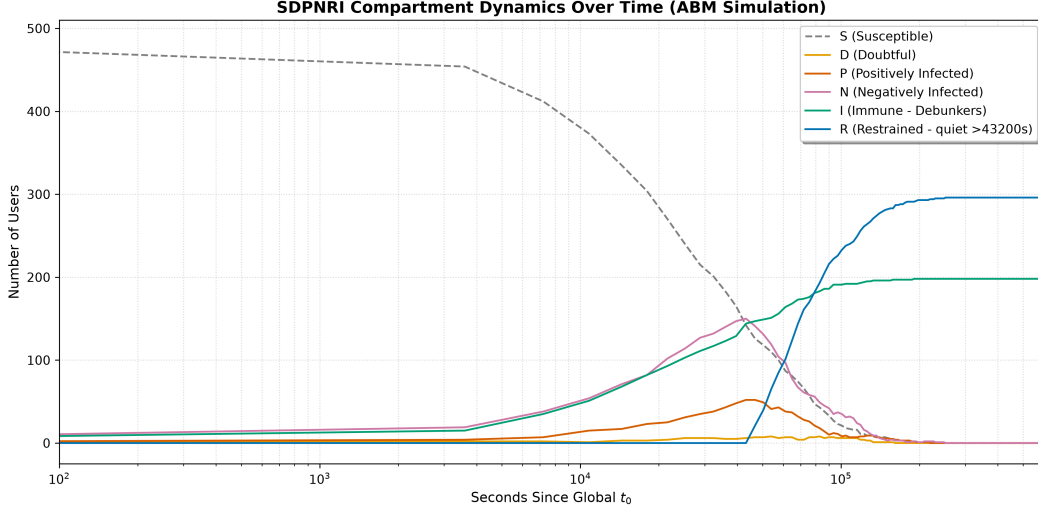


Figure 3: SDPNRI compartment dynamics over time from the Agent-Based Model simulation, compared against the empirical Boston Marathon bombings data.

extremes (Iyengar et al., 2019; Bail et al., 2018), this effective emotional activation ($E_{\text{effective}}$) is scaled by an emotional sensitivity exponent e , while the overall susceptibility to state transitions across network agents is modulated by the polarization sensitivity exponent b .

$$E_{\text{effective}} = (E_{\text{base}} \times (1 + |\text{Bias} - 1.0|))^e$$

Where:

- $\text{Bias} \in (0, 2]$ represents the agent's political ideology (1.0 being strictly neutral).
- e dictates how non-linearly emotional intensity scales transition rates.

To determine whether the user is inclined to spread the rumour with positive or negative sentiments, unnormalized directional weights w_P and w_N were computed. The political bias sensitivity exponent (b) governs the strength of ideological confirmation bias, non-linearly scaling how strongly an agent's political position influences their engagement framing:

$$w_P = E_{\text{effective}} \times (\text{Bias})^b$$

$$w_N = E_{\text{effective}} \times \left(\frac{1}{\text{Bias}}\right)^b$$

Finally, the baseline available probability pool β_{PN} is partitioned proportionally between the two states, ensuring that $P(S \rightarrow P) + P(S \rightarrow N) \leq \beta_{PN}$, with

$$\beta_{PN} = \beta_0 \times E_{\text{effective}} \times (1 + |\text{Bias} - 1.0|)^b$$

and

$$\beta_0 = P(S \rightarrow N) + P(S \rightarrow P) \approx 1.606367 \times 10^{-5}$$

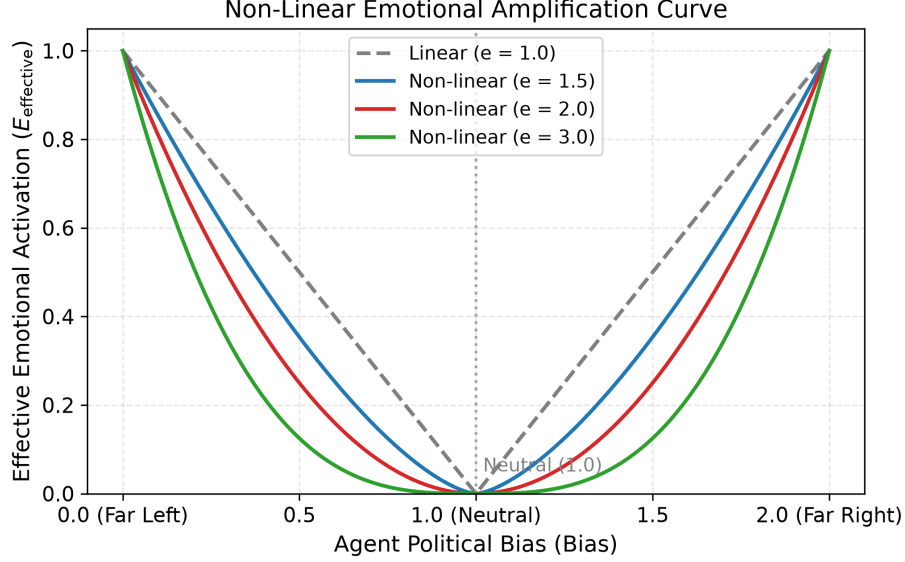


Figure 4: Non-Linear Emotional Amplification Curve.

$$P(S \rightarrow P) = \beta_{PN} \times \left(\frac{w_P}{w_P + w_N} \right)$$

$$P(S \rightarrow N) = \beta_{PN} \times \left(\frac{w_N}{w_P + w_N} \right)$$

3 Results

To address my research question, my ABM will simulate rumor propagation across a population of $N = 482$ agents over a 72-hour period. Agents are initialized with randomized political bias values drawn from a bi-modal Gaussian distribution ($\mu_1 = 0.4, \mu_2 = 1.6$) and uniformly distributed emotional intensity values. At each hourly time step, susceptible agents (S) undergo probabilistic transitions based on their emotional activation ($E_{effective}$) and political polarization exponent (b). To evaluate how political bias affects the threshold of Immune/Debunkers required to contain rumor outbreaks, I implemented an automated threshold-searching algorithm. First, the simulation sweeps across different values of the Susceptible to Immune transmission rate (p_{si}) with zero political bias ($b = 0$). The peak rumor attack percentage, defined as the proportion of infected individuals both negative and positive at the empirical baseline rate ($p_{si} = 0.03836$), that I calibrated using my data under $b = 0$, following the assumption that my selected data for my rumour has 0 political bias, is established as the target maximum outbreak size to avoid further spread of the rumour. For each elevated polarization level ($b \in \{0.5, 1.0, 2.0\}$), the sweep results are sorted in ascending order by debunker rate. The algorithm identifies the interval where the peak rumor attack crosses downward across the baseline target peak. we use linear interpolation to precisely calculate the required debunker threshold (p_{si}^*) when a crossover occurs between adjacent discrete test points ($y[i]$ and $y[i + 1]$).

$$\text{threshold} = \text{np.interp}(\text{baseline_peak}, [y[i + 1], y[i]], [x[i + 1], x[i]])$$

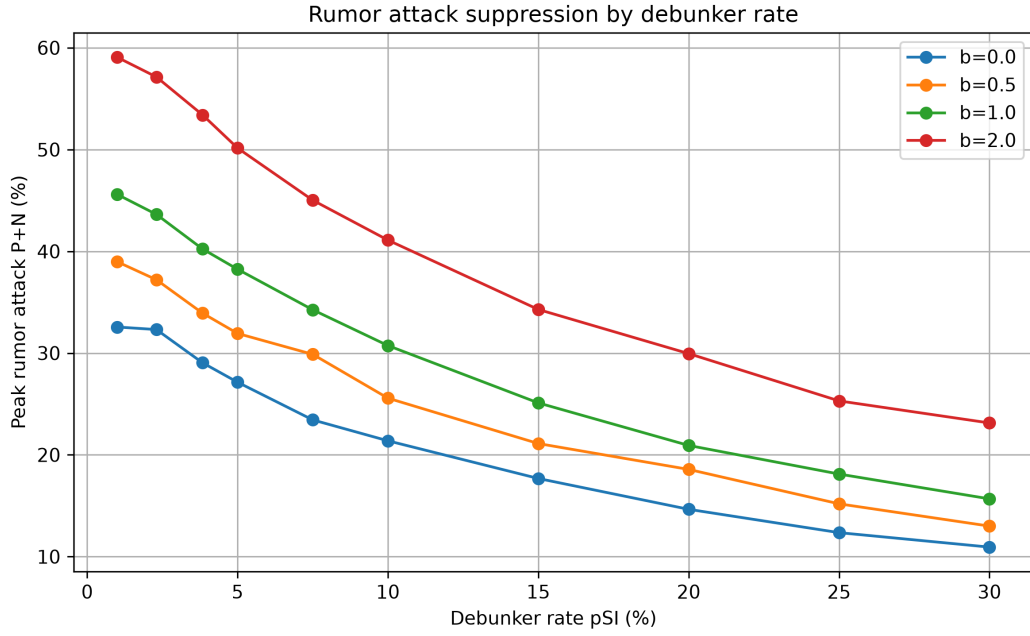


Figure 5: Rumor Peak Attack Rate vs. Debunker Rate (p_{SI}) with $R_0 \leq 1.0$ Threshold.

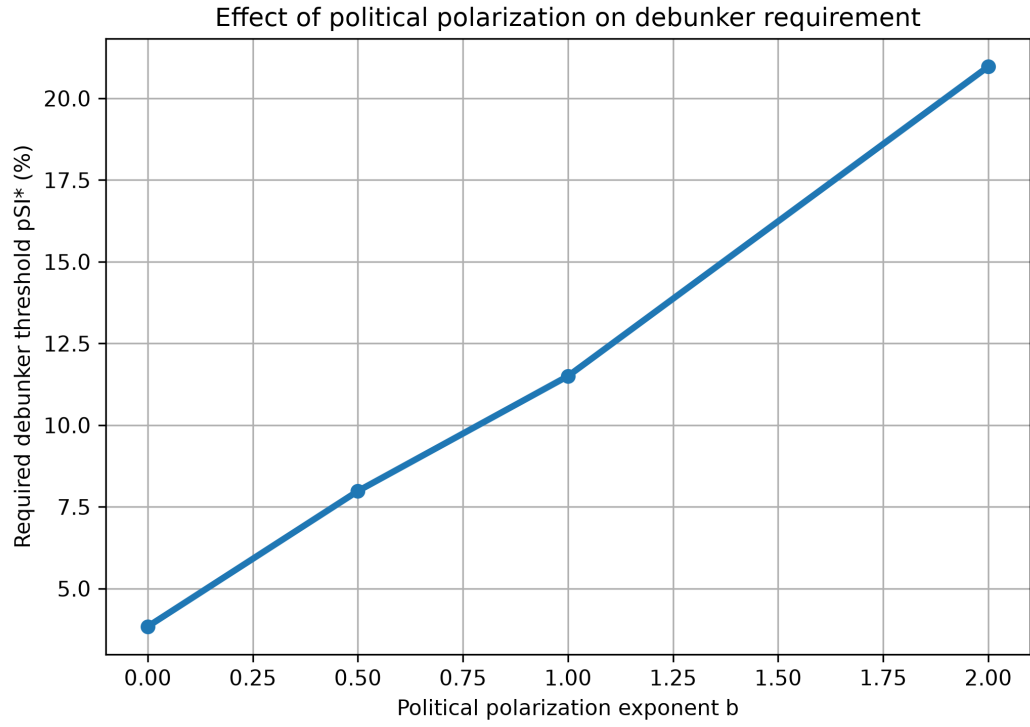


Figure 6: 72-Hour SDPNRI Compartment Dynamics across Polarization Levels (Debunker Rate $p_{SI} = 3.8\%$).

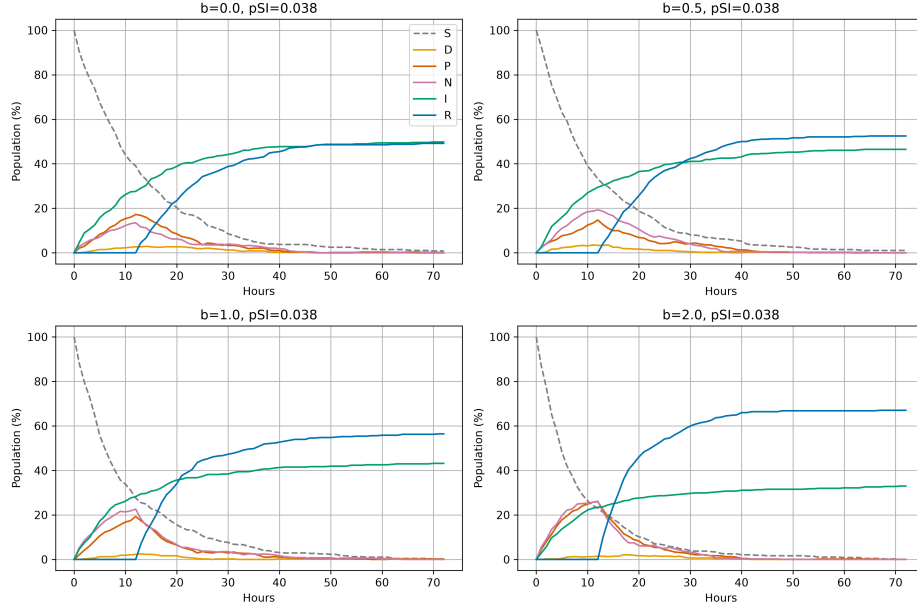


Figure 7: Critical Debunker Threshold (l^*) as a Function of Political Bias (b).

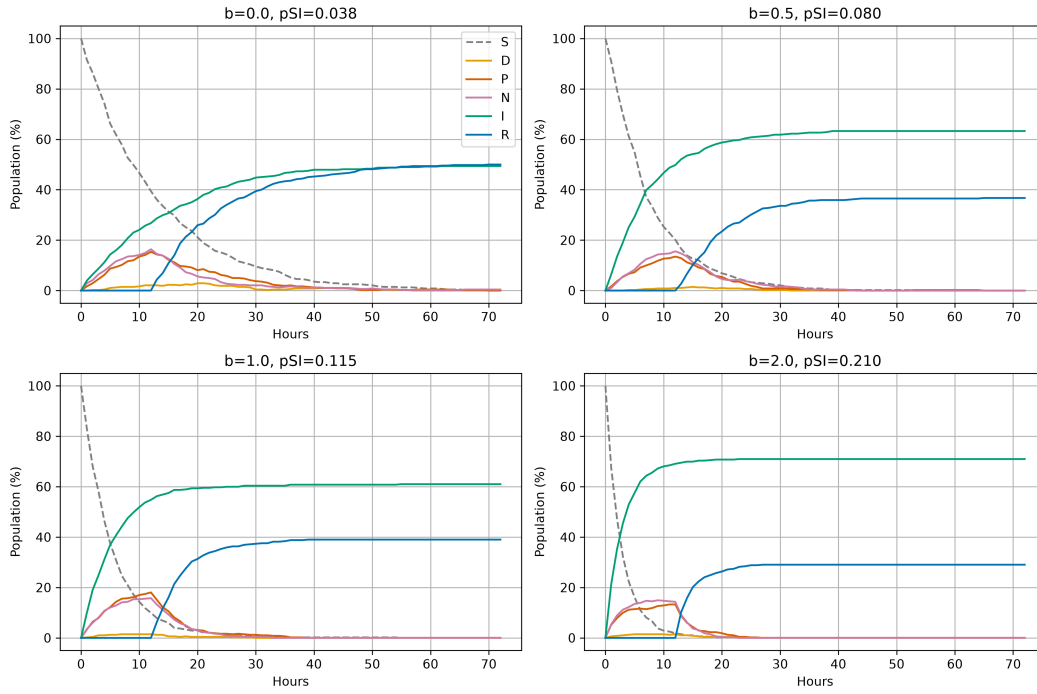


Figure 8: 72-Hour SDPNRI Compartment Dynamics at Theoretical $R_0 \leq 1.0$ Debunker Intervention Rates.

4 Findings

The simulation results demonstrate that political polarization significantly amplifies rumor propagation, shifting the peak attack curves upward across all debunker rates. When intervention capacity is held static at baseline levels, higher polarization leads to more intense, sharper rumor outbreaks and reduced overall immunity. However, by implementing an automated threshold-searching protocol, the model successfully determines that scaling the debunker transition rate (p_{si}^*) from 3.8% to over 21% per hour as polarization increases (b from 0.0 to 2.0) effectively locks and suppresses the peak rumor outbreak to match the unpolarized control level. Ultimately, this reveals a strong, nearly linear relationship establishing that heavily polarized populations require a proportionally heavier institutional intervention capacity to maintain control over misinformation. However, these results might seem a little bit unrealistic which will be discussed further in the limitation section.

5 Limitations:

Sample Size and Generalizability: The agent population size ($N = 482$) represents a relatively small sample, and baseline parameters are calibrated from specific empirical data (PHEME), which may limit the generalizability of absolute numerical values to broader, real-world social environments.

Quantification of Political Bias: The model operationalizes political polarization via a global exponent (b) and a simplified spatial distance metric relative to a neutral center, without establishing a dynamic mechanism to fully quantify or evolve individual political bias over time.

Dominance of the Susceptible-to-Immune Pathway: The $S \rightarrow I$ transition served as the primary and dominant mechanism for rumor suppression, while alternative corrective pathways ($D \rightarrow I$, $P \rightarrow I$, and $N \rightarrow I$) were barely existent in practice, which justified focusing optimization and threshold analysis exclusively on p_{si} .

6 Conclusion:

In conclusion, this study successfully demonstrates that while political polarization accelerates and intensifies rumor propagation, an automated threshold-based intervention protocol can effectively neutralize it. By dynamically scaling the debunker transition rate (p_{si}^*) from 3.8% to over 21% per hour as polarization increases (b from 0.0 to 2.0), the model successfully contains peak rumor outbreaks back down to baseline control levels. These findings highlight a strong, nearly linear relationship establishing that heavily polarized populations require a proportionally heavier institutional intervention capacity to maintain control over misinformation.

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